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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2025

Code: 105616 | ***Time: 03 Hours***

Semester: VI | ***Subject: Cryptography and Network Security*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) A polyalphabetic cipher can be classified as
- (i) Simple substitution
- (ii) Transposition cipher
- (iii) Multi-key substitution
- (iv) Stream cipher

(b) In transposition cipher, encryption is achieved by
- (i) Replacing characters with other characters
- (ii) Rearranging the order of characters
- (iii) Substituting binary digits
- (iv) Hiding message within images

(c) The key length of DES
- (i) 56 bits
- (ii) 128 bits
- (iii) 64 bits
- (iv) 256 bits

(d) The avalanche effect in DES ensures
- (i) Minimal change in ciphertext for key variation
- (ii) Key independence
- (iii) Constant ciphertext length
- (iv) Large change in ciphertext for small plaintext/key change

(e) Which of the following statements about finite fields is true?
- (i) Every finite field has a prime number of elements
- (ii) Finite fields exist only for even p^n
- (iii) Every finite field has p^n elements, where p is prime
- (iv) All finite fields are of infinite order

(f) If gcd(a, n) = 1, the modular inverse of a (mod n) exists and can be found using
- (i) Fermat's theorem
- (ii) Chinese remainder theorem
- (iii) Discrete logarithm
- (iv) Extended Euclidean algorithm

(g) Which of the following is a stream cipher?
- (i) RC4
- (ii) DES
- (iii) AES
- (iv) Triple DES

(h) The security of Triple DES relies on
- (i) Using three distinct keys with DES encryption-decryption-encryption
- (ii) XOR operations
- (iii) Key scheduling randomness
- (iv) Increasing block size

(i) The Chinese Remainder Theorem helps in
- (i) Finding public key
- (ii) Generating random numbers
- (iii) Solving modular equations efficiently
- (iv) Detecting prime numbers

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(j) Given p = 3, q = 11, public key e = 7, and plaintext M = 5, find ciphertext C.

(i) 12 (iii) 24

(ii) 14 (iv) 10

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain substitution and transposition cipher. Differentiate between both the techniques.

(b) Draw a neat diagram OSI Model. Discuss the various services provided by the OSI Security Architecture and how they interrelate.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Describe the structure and working process of DES with a neat diagram.

(b) Explain the concepts of differential and linear cryptanalysis. Explain Block Cipher Principles.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain Groups, Rings, and Fields. Use Euclid's Algorithm to find gcd(60, 24).

(b) Explain the role of key distribution and random number generation in ensuring confidentiality in network security.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Define AES. Explain the structure and major steps of the AES cipher.

(b) Explain the concept of block cipher modes of operation such as ECB and CBC with diagram.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Discuss in detail the RSA algorithm, including encryption, decryption, and security considerations.

(b) Explain Principles of Public-Key Cryptosystems. Explain Fermat's and Euler's Theorems with example.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain in detail the digital signature process. Explain the requirements for authentication in communication systems.

(b) Define message authentication. Differentiate between Message Authentication Code (MAC) and Hash Function.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Explain Playfair Cipher. A message "INFORMATION" is encrypted using a Playfair cipher with the keyword "KEYWORD". Construct the matrix and encrypt the message.

(b) Define confusion and diffusion. What is the purpose of hash functions in authentication systems?

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Differentiate between symmetric and asymmetric encryption. What is the purpose of hash functions in authentication systems?

(b) What is the difference between encryption and authentication? Find the multiplicative inverse of 7 mod 26.

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